

Computational Instinct Discovery

Simulation & Modeling Society

Welcome Candidate

This is not a test of your math or coding skills, but a look into your **computational instinct**. Please provide your reasoning.

Q1 The Coin Flip

You flip a fair coin and get heads 10 times in a row. On the 11th flip, is heads more likely, less likely, or equally likely? Explain your reasoning.

Q2 The Feedback Loop

If an app shows you more of what you "like," what happens to your feed over time? Is this effect generally beneficial or harmful, and why?

Q3 The Classifier Trap

A medical test that diagnoses a specific cancer returns "Positive" for every patient tested; it never yields a false negative. Why might this approach be fundamentally unsuitable for medical diagnosis?

Q4 Growth Dynamics

A meme gets a few likes. More people see it → more people like it → even more people see it. Describe the phenomenon happening here. Can this growth continue indefinitely? Why or why not?

Q5 The Cost of Experimentation

A car manufacturer wants to test if a new car model can safely survive a 50 mph crash. Why shouldn't they test this by crashing real cars repeatedly? In your answer, touch upon the role of simulation.

Thank you

Thank you for your insights. Your responses will help the Committee understand your unique approach to problem-solving.